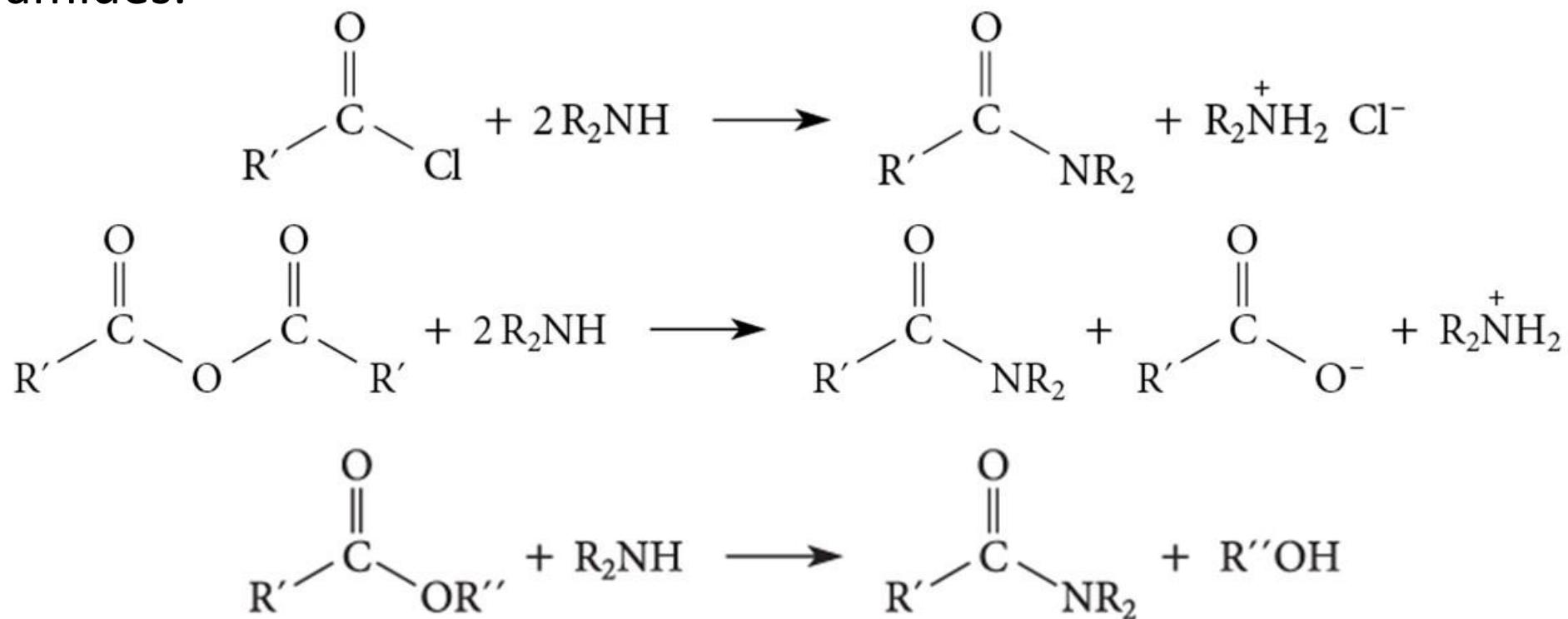


Chapter 23 (Carey Chapter 22)

The Chemistry of Amines

Acylation of Amines

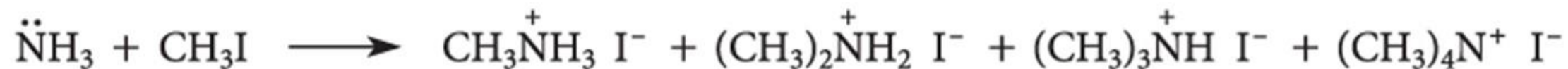
- Acid chlorides, anhydrides, and esters can react with amines to make amides.



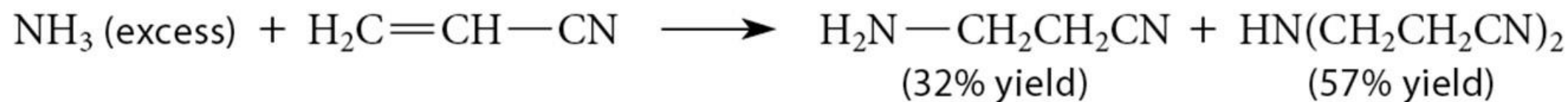
Direct Alkylation of Amines with Halides



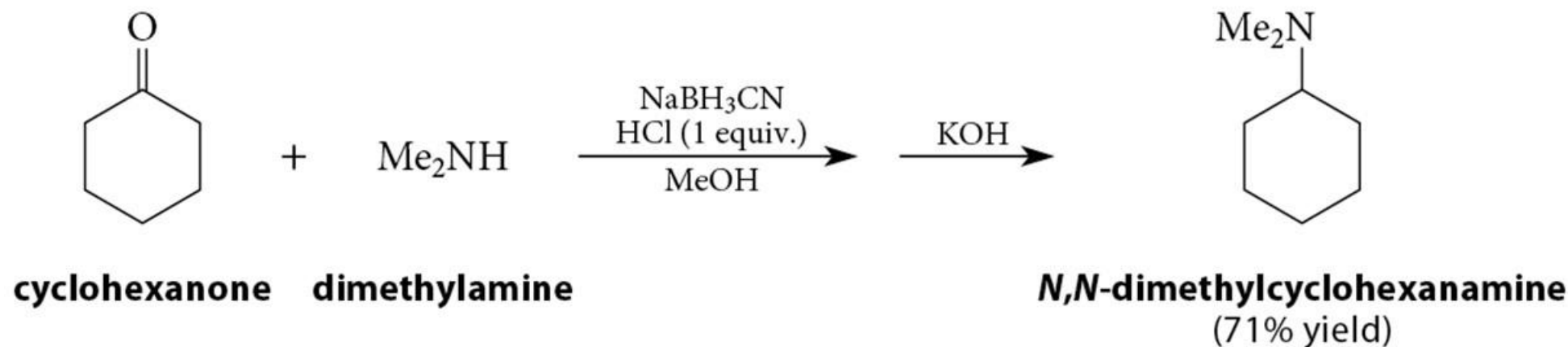
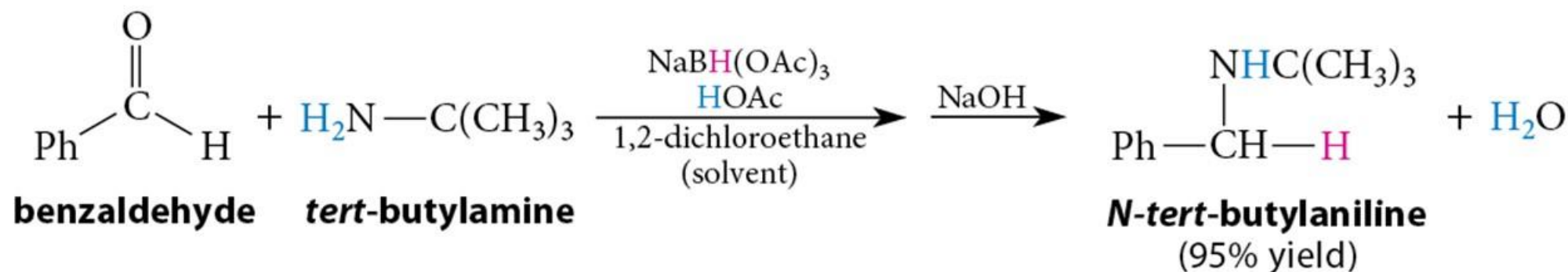
- Further alkylations can take place to give complex mixtures:



Alkylation of Amines: Epoxides and α,β -Unsaturated Carbonyls and Nitriles

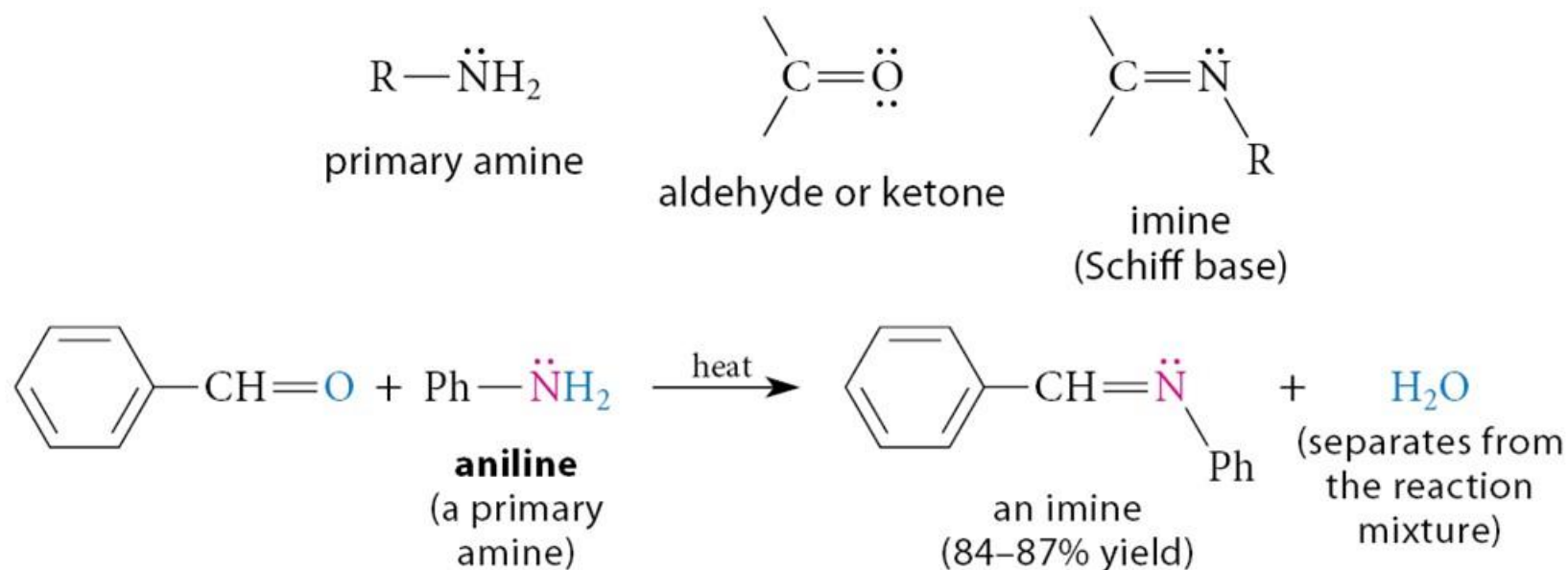


Reductive Amination: Hydride Reducing Agents



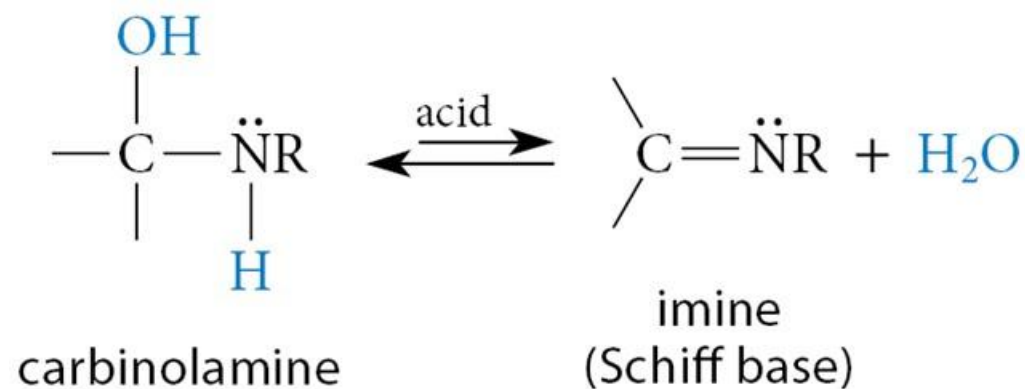
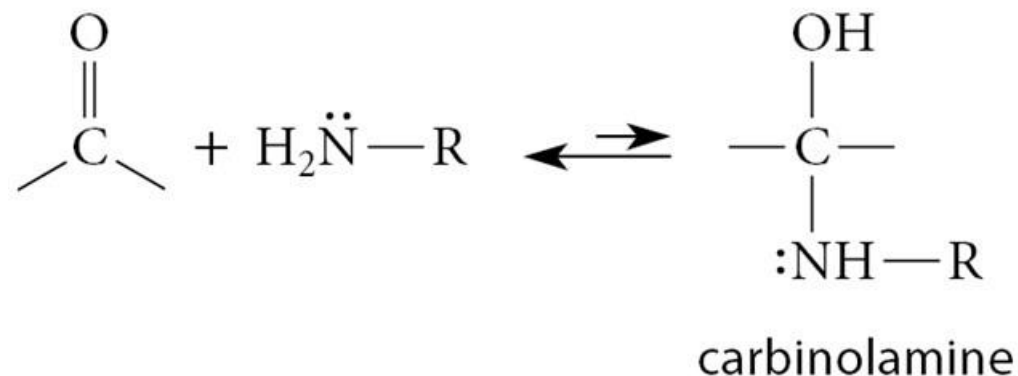
Recap: Reactions with Primary Amines

- Reaction of a primary amine with an aldehyde or ketone provides an imine product (also called a **Schiff base** or a **Schiff's base**).

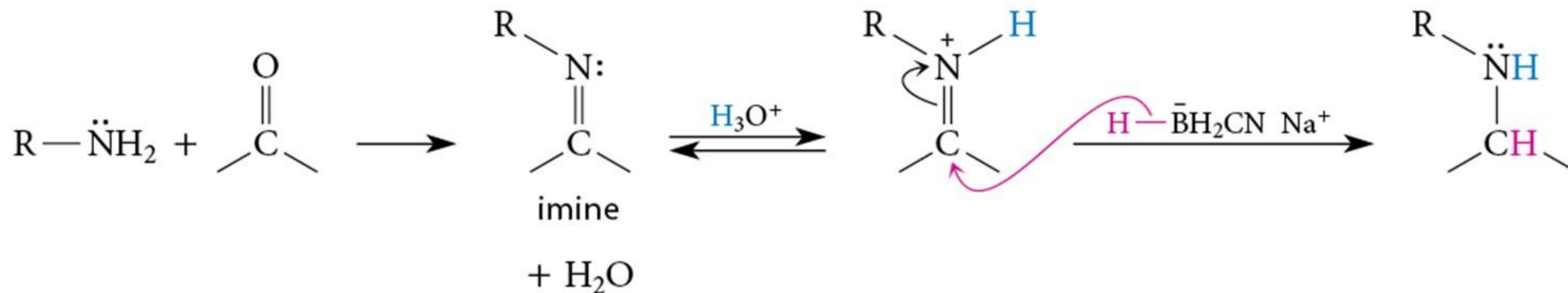


Recap: Mechanism of Reaction with Primary Amines

- Formation of carbinolamine
- The carbinolamine dehydration is typically the rate-limiting step.

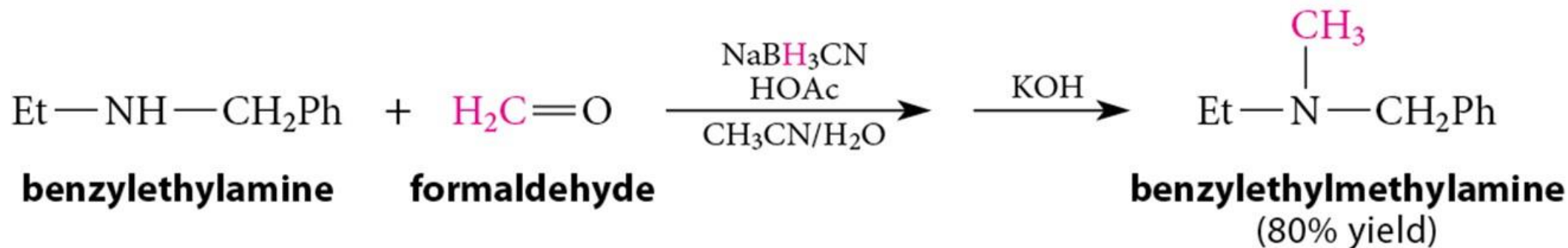
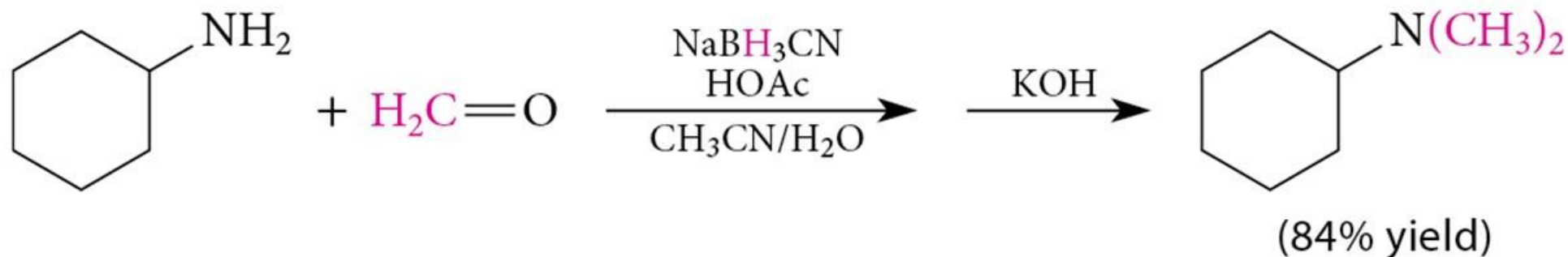


Reductive Amination: Imine Intermediate



Reductive Amination: Introducing Methyl Groups

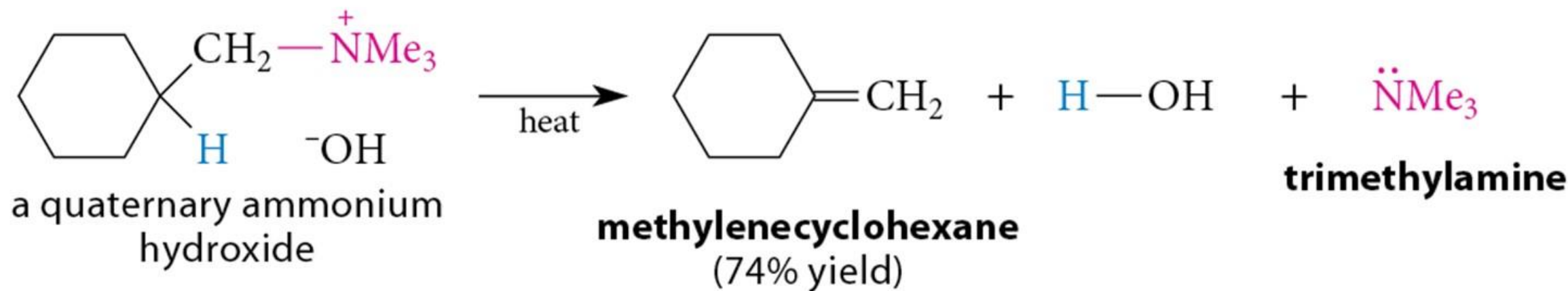
- Formaldehyde can be used to introduce methyl groups



- Comparison of borohydride reagents on the board.

The Hofmann Elimination

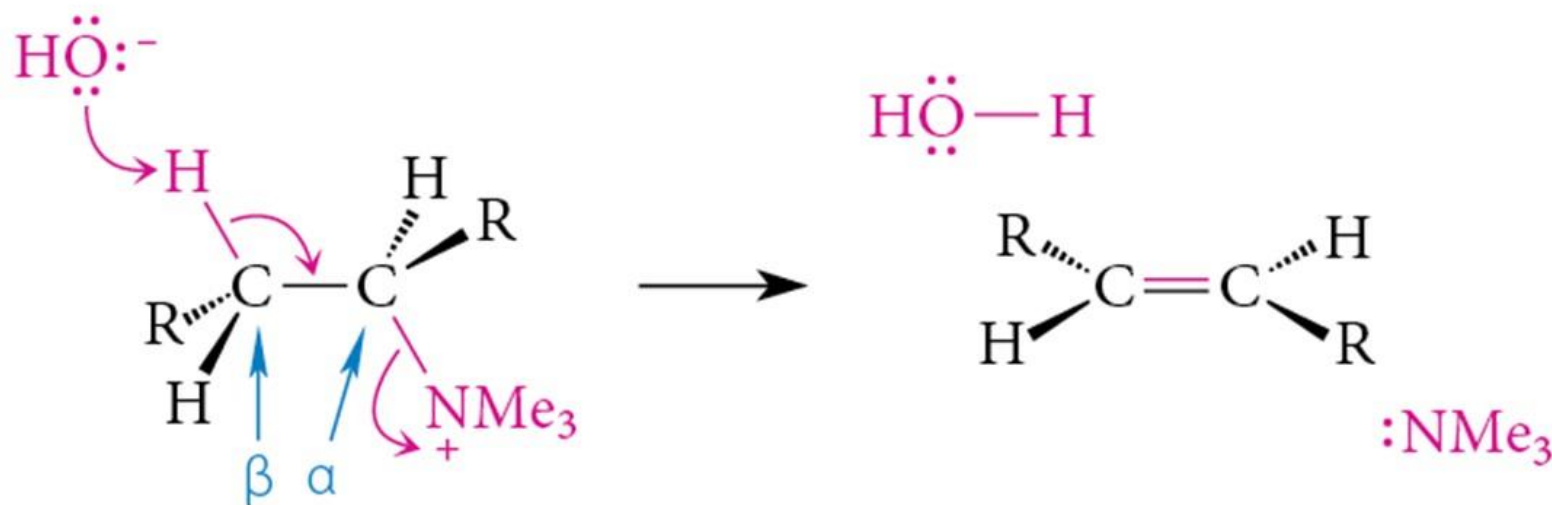
- When a quaternary ammonium hydroxide is heated, a β -elimination reaction takes place to give an alkene.



23.8 Hofmann Elimination of Quaternary Ammonium Hydroxides

The Hofmann Elimination: Mechanism

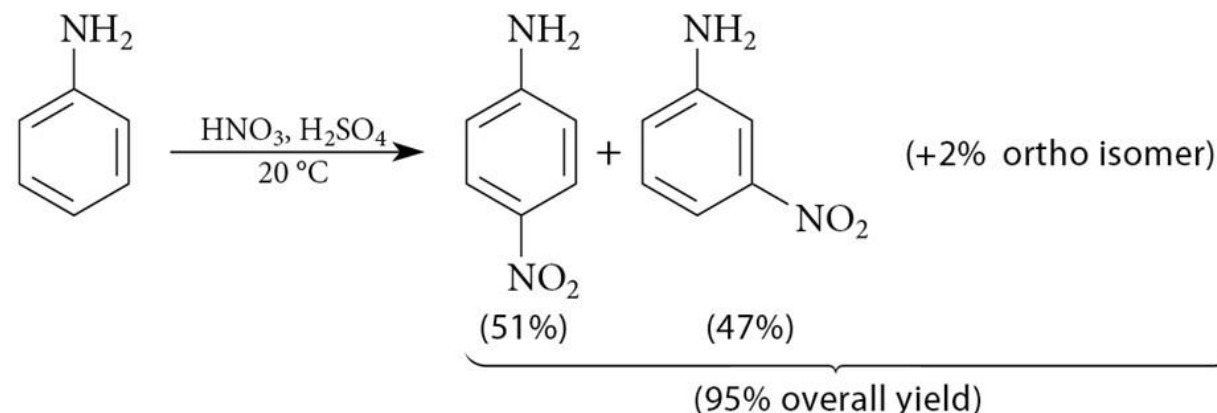
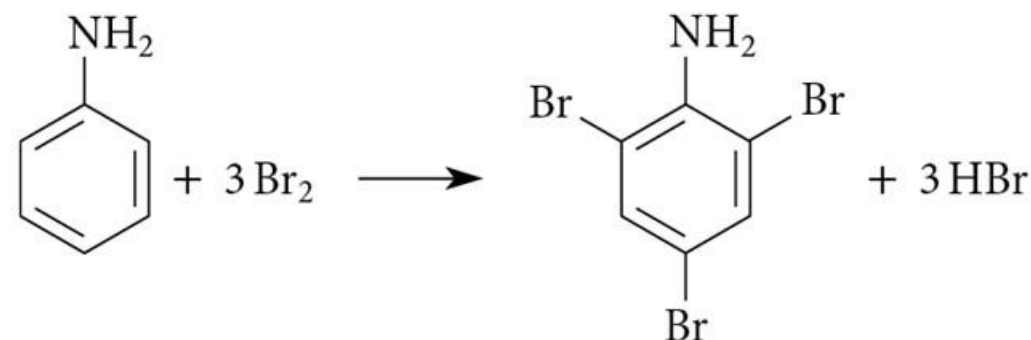
- Like the E2 reaction of alkyl halides, the Hofmann elimination is generally an anti-elimination.



23.8 Hofmann Elimination of Quaternary Ammonium Hydroxides

Aromatic Substitution: Aniline Derivatives

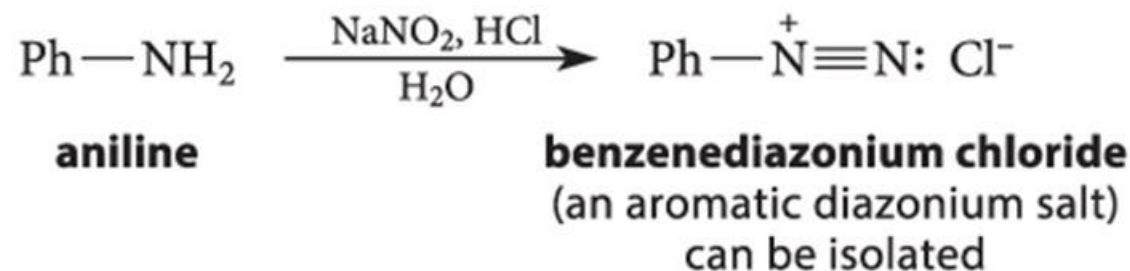
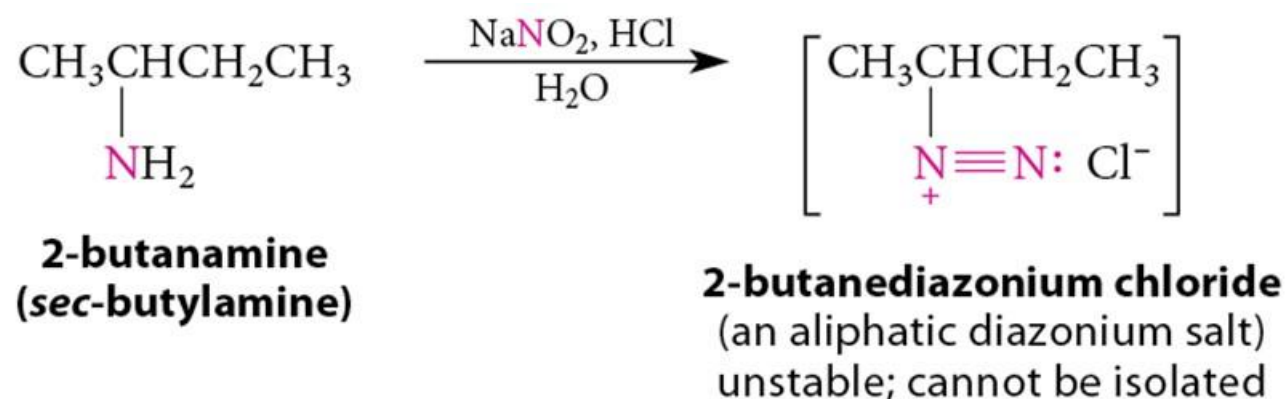
- The amino group is a powerful activating group that directs *ortho* and *para*.
- When treated with nitric and sulfuric acid, both *meta* and *para* nitration is observed.



Similarity of “enamine chemistry” and “enolate chemistry” on the board.

Formation of Diazonium Salts

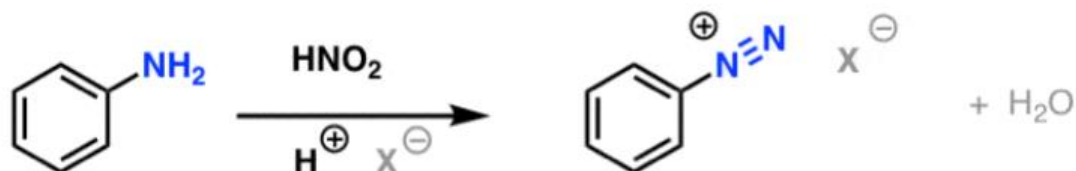
- **Diazotization:** The reaction of primary amines with nitrous acid (HNO_2).



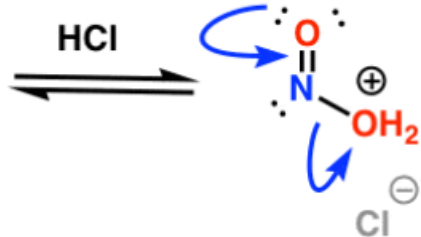
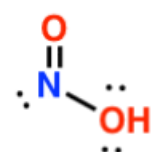
23.10 Diazotization; Reactions of Diazonium Ions

Formation of Diazonium Salts

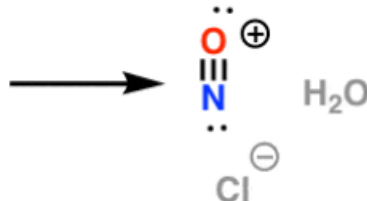
Formation of Diazonium Salts From Aromatic Amines



Nitrous acid



Nitrosonium ion

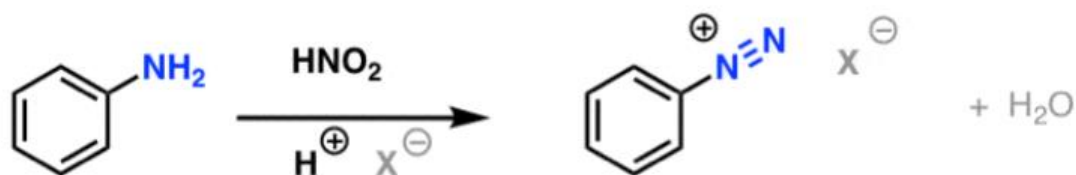


<https://www.masterorganicchemistry.com/2018/12/03/reactions-of-diazonium-salts-sandmeyer-and-related-reactions/>

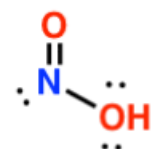
23.10 Diazotization; Reactions of Diazonium Ions

Formation of Diazonium Salts

Formation of Diazonium Salts From Aromatic Amines



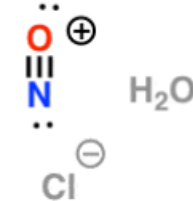
Nitrous acid



HCl

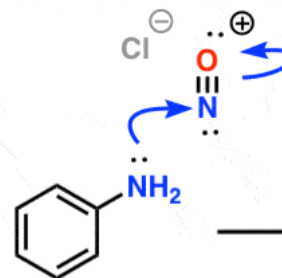


Nitrosonium ion

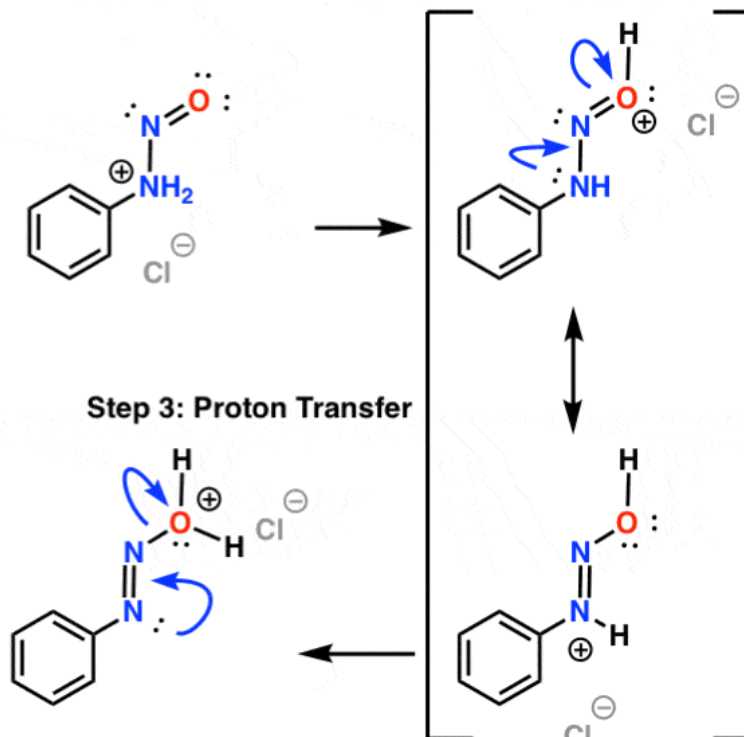


Mechanism: Formation of Diazonium Ions From Aromatic Amines

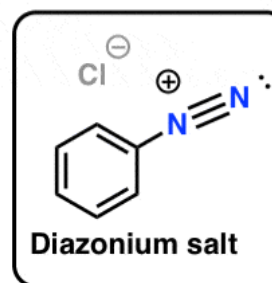
Step 1: Addition to nitrosonium ion



Step 2: Proton Transfer



Step 4: Elimination of H₂O



Step 3: Proton Transfer

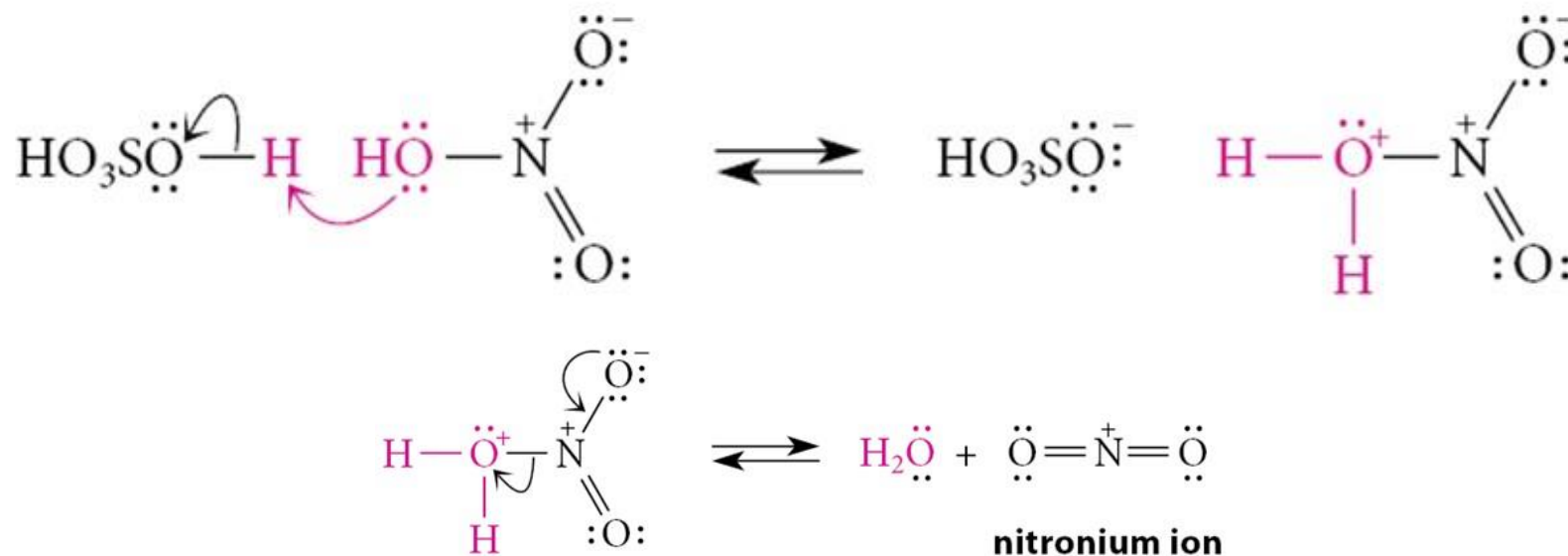
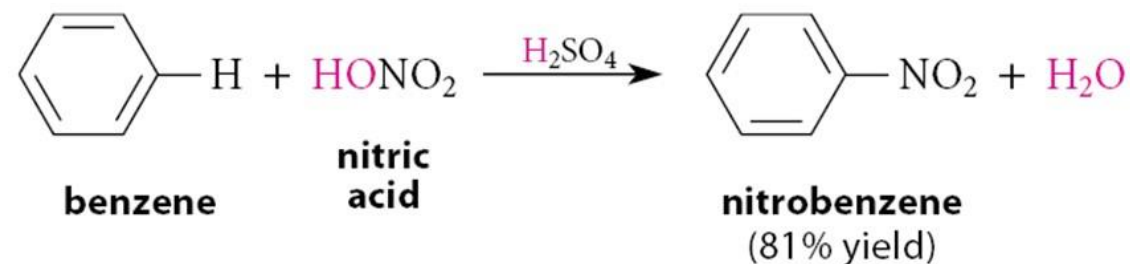
+ H₂O

<https://www.masterorganicchemistry.com/2018/12/03/reactions-of-diazonium-salts-sandmeyer-and-related-reactions/>

23.10 Diazotization; Reactions of Diazonium Ions

Recap: Nitration of Benzene (1 of 2)

- Overall Reaction
- Mechanism
 1. Generation of the electrophile:

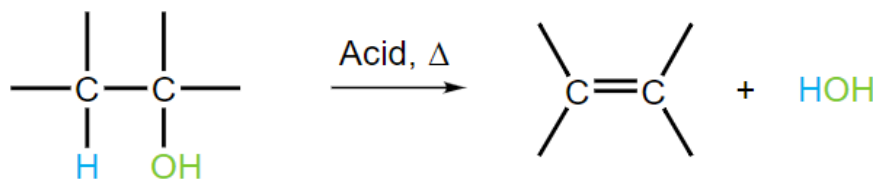


Recap: Dehydration reaction

- Acid causes elimination of water from OH compounds

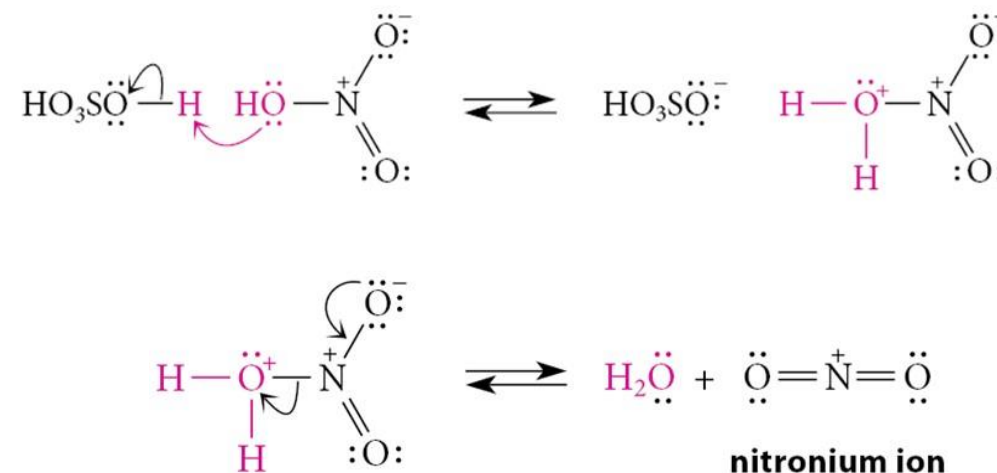


Alcohol dehydration:



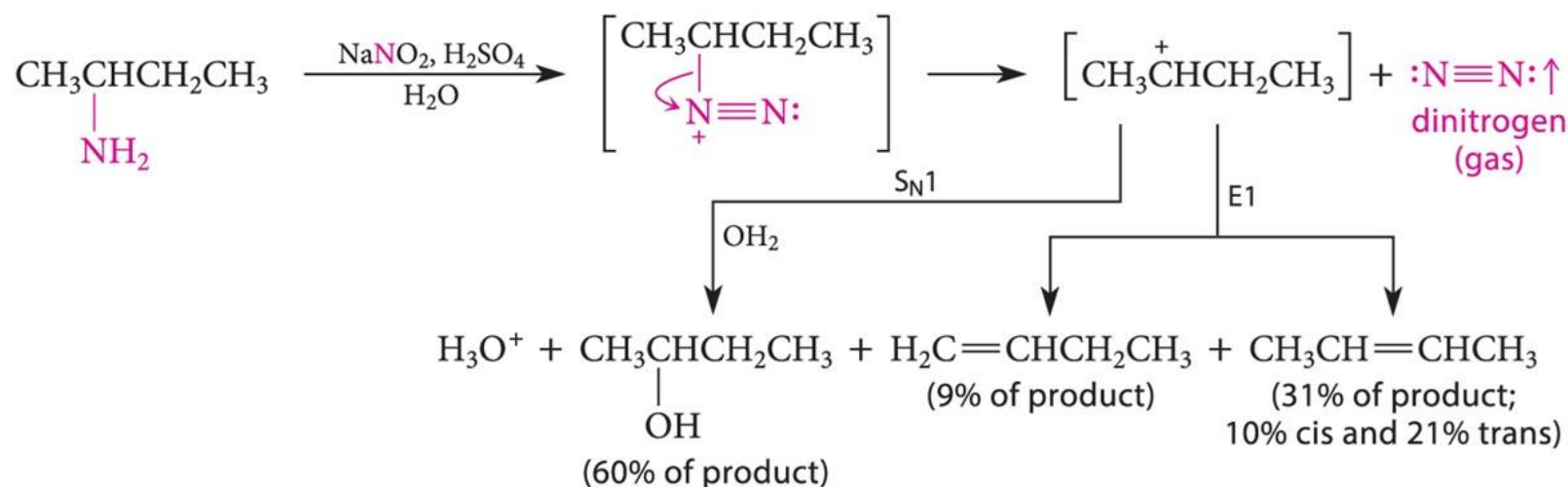
Order of Reactivity: primary < secondary < tertiary

Major product: thermodynamic product



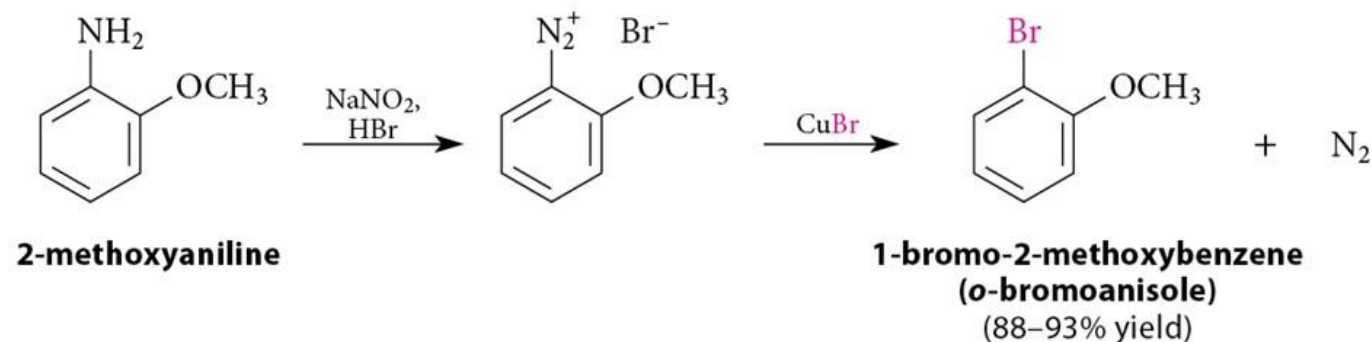
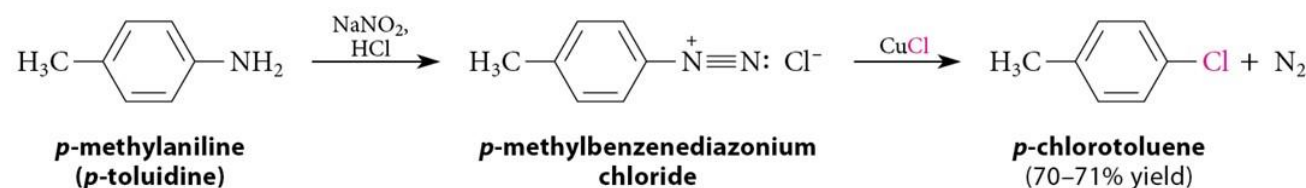
Reactions of Aliphatic Diazonium Salts

- Diazonium salts possess an *exceptional* leaving group: $\text{N}\equiv\text{N}$
- Aliphatic diazonium salts react immediately to give a complex mixture of products.



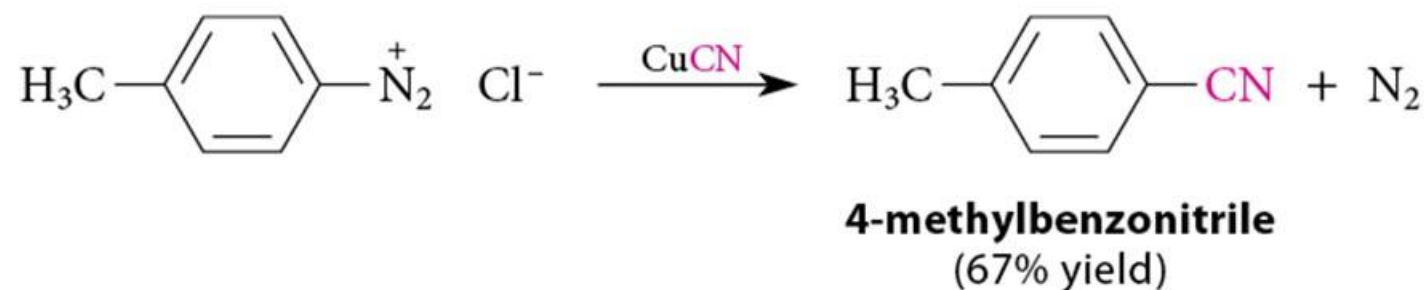
Reactions of Aryldiazonium Salts (1 of 3)

- Aryldiazonium salts are synthetically much more useful.
- The reaction of an aryldiazonium salt with a cuprous salt is called the **Sandmeyer reaction**

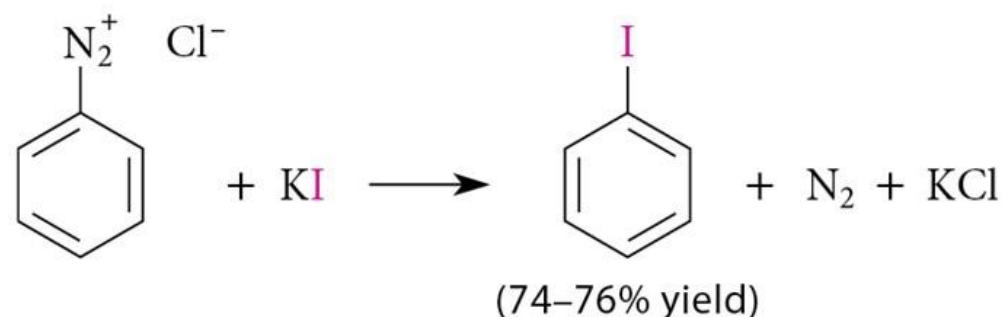


Reactions of Aryldiazonium Salts (2 of 3)

- The diazonium group can be replaced with a cyano group using CuCN.

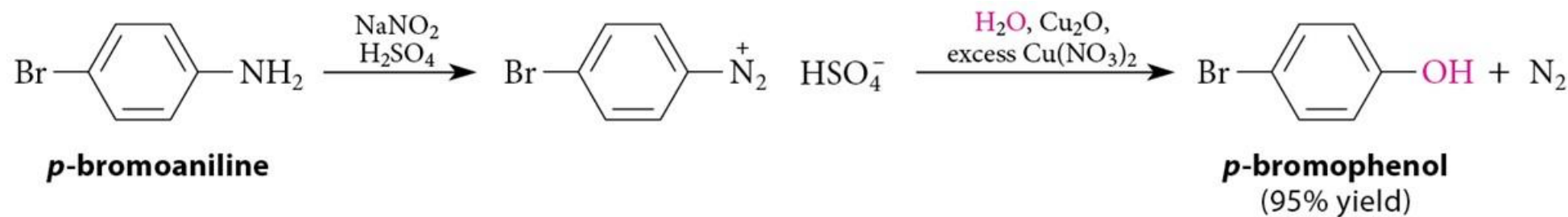


- Aryl iodides can be prepared by reacting the diazonium salt with KI.



Reactions of Aryldiazonium Salts (3 of 3)

- Hydrolysis to phenols:



- Replacement by hydrogen:

